7090 System Course

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Text: "Reference Manual IBM 7090 Data Processing System"
Form No. A22-6528-3 (August 1961)

References: IBM Customer Engineering Manual of Instruction
7090 Data Processing System, Vol. 1 and Vol. 2
Form No. 223-6860-1 (March 1960)

Lecture I. Introduction (Review of Computer Fundamentals)
Reference: Pages 7, 115-119

(1) Review concepts of system design
(2) The 5 parts of a computer
(3) Serial and parallel arithmetic
(4) Number bases--binary, octal
(5) Addressing--single address logic
(6) Registers--name the 7090 registers
(7) Review history of 7090--701, 704, 709, 7040, 7094

Main Homework Assignment for Course: Make notebook listing the format and a few descriptive words for each instruction in the 7090. Hand in pages as completed.

Lecture II. Word Formats and Addressing

Reference: Pages 8-12

(1) Fixed-point numbers
binary point
signs
1's complements and 2's complements

(2) Floating-point numbers
characteristic--range
fraction--binary point

Machine registers

Accumulator
MQ
Instruction register, address register, location counter

(3) Instruction formats
prefix, decrement, tag, address

Explain operation of index registers--complements
$A_{eff} = A - C(XI)$

(4) Addressing
Immediate
Direct--regular and indexed
Indirect

(5) Indicators, switches and lights

Lecture III. Fixed Point Arithmetic--Add-type instructions

Reference: Pages 18-21, 96-99

(1) Review addressing. Describe FAP notation briefly.
Work out examples of adding complements, etc.

(2) Explain "ORing" of multiple index registers

(3) Explain timing cycles in Instruction fetching and execution
I cycles
E cycles
L cycles

(4) Start through the instructions

CLA	Explain data flow
CAL	Explain difference between logical word and numerical word
CLS	
ADD	Go through binary addition truth table, etc.
ADM	Explain 4 cases: add, sub; abs, algebraic.
SUB	
SBM	
ACL	

Lecture IV. Fixed Point Arithmetic--Multiply and Divide

Reference: Pages 22-25

(1) MPY - describe meaning of double length product
Explain how adders and shifters work.
Go through example in book in detail

(2) RND- need for round

(3) VLM - explain in terms of MPY cycles

(4) DVH - explain double-length dividend
Go through example in book in detail--testing necessary
Explain "Halt" case

(5) DVP

(6) VDH and VDP - explain in term of DVH cycles

Lecture V. Floating Point Arithmetic

Reference: Pages 25 - 31

(1) Review floating point data format
Explain range of numbers--the problem with overflows and underflows

(2) FAD - Go through in great detail each of cases--signs, zeros, etc.

(3) FAM - Explain 8 cases: add, sub; norm, unnorm; abs, algebraic,
UFA
FSD
UAM
FSM
UFS
UFM

(4) FMP - Go through example
UFM
FRN

(5) FDH - Explain difference from fixed point cases
FDP

(6) Floating point trap operation

Lecture VI. Shifting, Word Transmission, and Control Instructions

Reference: Pages 31 - 39

(1) Review registers - Acc, MQ, etc.
Review "logical word"

(2) ALS LRS LGL RQL
ARS LLS LGR

Explain operation of each.
Explain "packing and unpacking"

(3) Loads and stores
Discuss fields and words. Go through each briefly:

LDQ	STD	STP	STA	STL	XCA	ENK
STQ	SLW	STD	STZ	STR	XCL	
SLQ		STT				

(5) Control ops

NOP
HPR
HTR - Review meaning of branch
TRA

(6) Trapping Mode - explain

ETM
LTM
TTR

(7) Conditional arithmetic branches

TZE	TPL	TOV	TQP	TQO	TLQ
TNZ	TMI	TNO			

Explain 8 possible cases of branch: Tr, notr; no, off; leave, reset

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Lecture VII. Index Branches, Sense Branches, and Index Transmission

Reference: Pages 39-43, 44-47

(1) TSX

Explain subroutine entry and return. loops

(2) TXI
TXH
TXL

Explain three operations to be done:

(1) step up index
(2) test condition
(3) branch

(3) TIX
TNX

Explain combined operation--give example

(4) Sense type branch instructions

PSE
MSE
BTT
ETT
IOT
PBT
LBT
DCT

Explain each briefly

(5) Arithmetic sense-type tests

ZET
NZT
CAS
LAS

(6) Index transmission operations

Purpose: to permit arithmetic on index registers, and use of more indices

LXA	LXD	PAX	PDX	SXA	PXA
LAC	LDC	PAC	PDC		
AXT - immediate type				SXD	PXD
AXC					

Lecture VIII: Logical Operations, Sense Indicator Operations, and Comments

Reference: Pages 47 - 57

(1) Logical Operations

Review logical connectives
16 plus "to acc" or "to storage"-- 32 possible

ORA	ANA	ERA
ORS	ANS	

(2) Sign Controls

COM	complement--leave sign
CLM	clear-- " "
CHS	change
SSP	plus
SSM	minus

(3) Indicator Operations

PAI	LDI	place, load and store--cover each briefly
PIA	STI	
OAI	"or"accumulator	
OSI		

SIL	Set indicators of left half			
SIR				
RIA	Reset		IIA	Invert
RIS			IIS	
RIL			IIL	
RIR			IIR	
TIO	Transfer on			
TIF	Transfer off			
ONT	on	LNT	left	LFT
OFT	off	RNT	right	RFT

(4) Convert Instructions

Rapid table look-up
Go through examples pp 55 and 56

CVR	Convert by replacement from AC
CRQ	Convert by replacement from MQ
CAQ	Convert by addition from MQ

Lesson XI. I/O and Channel Operations (really needs 2 lectures)

Reference: Pages 12-17, 43-44, 57-65, 77-95

(1) Channel Registers

Count reg - decrement field 3 - 17
Loc. reg. --address field 21-35
Operation reg. - S, 1, 2, 19

Two addresses: Unit-channel and CPU (see p. 57)
Two programs: CPU and channel

Trapping signals

(1) completion of command
(2) EOF
(3) Tape check

(2) Testing channel conditions

TCOA	Transfer channel A in operation
TCNA	
TRCA	Transfer on redundancy (tape check) - reset
TEFA	Transfer on end of file
TCH	Channel program branch

(3) Input-Output Operations

RDS	Read
WRS	Write
BSR	Record
BSF	File
WEF	End of file
REW	Rewind
RUN	Rewind and unload
SDN	Set density - tape

(4) Channel Transmission Operations

RDCA	Reset channel
SCHA	Store channel (address, loc., op.)
RCHA	Reset and load
LCHA	Load

(5) Channel Commands

IOCD	Count control and disconnect
IOCP	Count control and proceed
IORP	Record (or count) and proceed
IOCT	Count control and transfer
IORT	Record (or count) and transfer
IOSP	To record or count and signal and proceed
IOST	To record or count and signal and transfer

Go through examples pp. 64-65 - stress two programs.

(6) Channel Trap Operations

ENB	Enable
RCT	Restore

Lesson X. Compatibility and Disk Instructions

Reference: Pages 65 - 72

(1) System Compatibility Operations

ESNT LSNM	Storage nullification mode
ESTM	Select trap mode
ECTM	Copy trap mode
EFTM LFTM	Floating Point Trap

Refer to p. 31

(2) Disk Storage Instructions

(a) CPU

RSCA	ENB
STCA	RICA
SCHA	

(b) I/O Commands

CTL
CTLR
CTLW
SNS
CPYD
CPYP

(c) Channel Interrupt and Control Commands

SMS	TCH	LIP	XMT
ICC	WTR	LAR	
	TWT	SAR	
	TCM	LCC	
	TDC		

Lesson XI. Review Material and Program Examples

Lesson XII. Final Exam